

Kavin Desi Valli

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TECHNICAL SKILLS

Languages: Rust, C++, TypeScript, Python, System Verilog, C++, SQL

Frameworks: React, Next.js, Node.js, OpenTelemetry, Kafka, Protobuf, gRPC, Redis, PostgreSQL, Django, Flutter

Other Tools: Docker, Git, OpenCV, AWS, Vercel, Cloudflare Workers

EXPERIENCE

Software Engineer

Remote

Replicas

Aug. 2026 – Present

- Designed and shipped infrastructure for **cloud coding agents**, including request triage, Slack integrations, and automated workspace provisioning.
- Built production integrations across **Slack**, **GitHub**, and **Linear** to automate agent workflows, PR lifecycle events, issue synchronization, and notifications.
- Improved product performance through dashboard refactors, client-side caching, prefetching, and reusable integration components.

Software Engineer Intern

San Francisco, CA

Vercel

Jan. 2026 – Apr. 2026

- Engineered a **dual-VM sandbox pooling system** that reduced infrastructure costs by **20×** while improving sandbox startup performance.
- Shipped the **v0 Slack Agent** to General Availability, contributing production-ready features and launch readiness.
- Customized and integrated a **fork of VS Code's code-server** into v0 to enable a browser-native AI coding environment.
- Developed production features and performance improvements across the v0 platform in a fast-paced engineering team.

Software Engineer Intern

San Francisco, CA

Helicone AI (YC W'23) 

Sept. 2024 – Dec. 2024, May 2025 – Aug. 2025

- Shipped new features and bug fixes in a rapid startup environment, merging approximately **200 pull requests**.
- Built production features using **Node.js**, **Next.js**, **Rust**, **OpenTelemetry**, **Protobuf**, **HDBSCAN clustering**, and **Kafka**.
- Created and maintained NPM, Python, and Go packages including [@helicone/async](https://github.com/helicone/async), [@helicone/helpers](https://github.com/helicone/helpers), [helicone-async](https://github.com/helicone-async), [helicone-helpers](https://github.com/helicone-helpers), and [go-helicone-helpers](https://github.com/go-helicone-helpers).
- Implemented features and reliability improvements in Helicone's Rust-based [AI Gateway](#).

Embedded Full Stack Developer

Toronto, Canada

Arcturus Networks Inc.

Jan. 2024 – Apr. 2024

- Independently developed the Brinq Edge AI platform for the **NXP i.MX 8M Plus** using **React**, **Node.js**, and **C++**, with development and testing performed on a **Jetson Xavier NX**.
- Integrated **YOLOv8 object detection**, **occupancy analytics**, and video clip saving using **FFmpeg**.
- Developed an **OpenCV** heatmap visualizing crowd intensity from real-time **HLS video streams**, showcased at **Embedded World 2024** on an **i.MX 93 SoM**.
- Reduced initial load times of **10–15 seconds**, optimized video storage using a circular buffer, and maintained deployment and **Jenkins pipelines**.

PROJECTS

Pipelined Matrix-Vector Accelerator | *SystemVerilog, Vivado, FPGA*

- Designed an **8-lane pipelined matrix-vector accelerator** in SystemVerilog, achieving **~400 MHz** on an AMD Kria KV260 FPGA using BRAM-backed memories and streaming interfaces.

Typewind | *TypeScript, Rust, Next.js*

- Developer and maintainer of an open-source project with **2.1K+ GitHub stars**, combining TypeScript's type safety with the versatility of Tailwind CSS. 

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Computer Engineering

Sept. 2023 – Apr. 2028

ACHIEVEMENTS

NASA Space Apps Global Finalist Honorable Mention | *NASA Earth Science Division*

Nov. 2022

- Ranked among the top **80 of 50,000+ teams** for building [Liberty](#).